MIL-M-38510/329A
22 August 1983
SUPERSEDING
MIL-M-38510/329(USAF)
18 December 1978

MILITARY SPECIFICATION

MICROCIRCUIT, DIGITAL, LOW-POWER SCHOTTKY TTL,

PARITY CHECKER, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, low-power Schottky TTL, parity checker logic microcircuit. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, and as specified herein.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	9-bit odd/even parity generator/checker

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
X	C-2 (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.5 V dc at -18 mA to +5.5 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/	148.5 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases A, B, and D	70°C/W
Case C	50°C/W
Case X	60°C/W
Junction temperature (T_J)	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage range (V_{CC})	+4.5 V dc minimum to +5.5 V dc maximum
Minimum high-level input voltage (V_{IH})	+2.0 V dc
Maximum low-level input voltage (V_{IL})	+0.7 V dc
Normalized fanout (each output)	10 maximum
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagrams. The terminal connections and logic diagrams shall be as specified on figures 1 and 2, respectively.

3.2.2 Truth tables. The truth tables shall be as specified on figure 3.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All manufacturers' schematics shall be maintained and available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = +4.5 V, I_{OH} = -.4 mA	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = +4.5 V, I_{OL} = +4.0 V		0.40	V
Input clamp voltage	V_{IC}	V_{CC} = +4.5 V, I_{IN} = -18 mA, T_C = +25°C		1.5	V
Low-level input current at inputs	I_{IL}	V_{CC} = +5.5 V, V_{IN} = +0.4 V	-120	-370	µA
High-level input current	I_{IH1}	V_{CC} = +5.5 V, V_{IN} = +2.7 V		20	µA
	I_{IH2}	V_{CC} = +5.5 V, V_{IN} = +5.5 V		100	µA
Short circuit output current	I_{OS}	V_{CC} = +5.5 V, V_{OUT} = GND 1/	-15	-100	mA
Supply current	I_{CC}	V_{CC} = +5.5 V, V_{IN} = GND		27	mA
Propagation delay time low-to-high level output from input to Σ even	t_{PLH1}	V_{CC} = +5.0 V, R_L = 2 kΩ ±5%, C_L = 50 pF ±10%	2	72	ns
Propagation delay time high-to-low level output from input to Σ even	t_{PHL1}		2	65	ns
Propagation delay time low-to-high level output from input to Σ odd	t_{PLH2}		2	52	ns
Propagation delay time high-to-low level output from input to Σ odd	t_{PHL2}		2	72	ns

1/ Not more than one output should be shorted at one time.

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3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking of the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 11 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7, 9,10,11	1*2,3,7, 9
Group A test requirements (method 5005)	1,2,3,7, 8,9,10,11	1,2,3,7, 9,10,11
Group B test requirements (method 5005) subgroup 5	1,2,3,7, 8,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	N/A
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

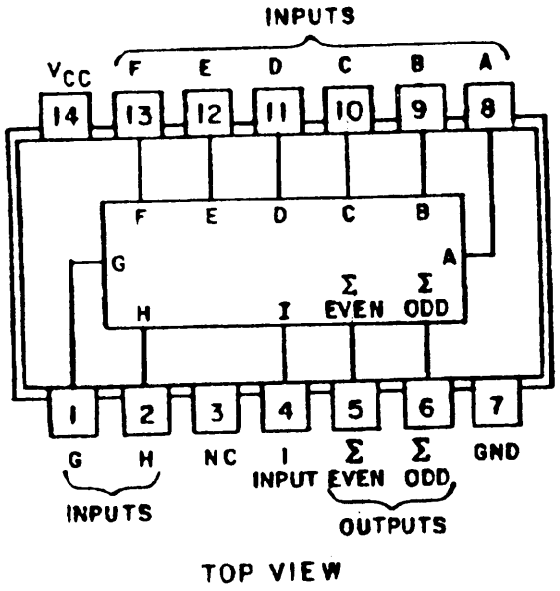
(2) T_A = +125°C minimum.

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Device type 01

Cases A, B, C and D



Device type 01

Case X

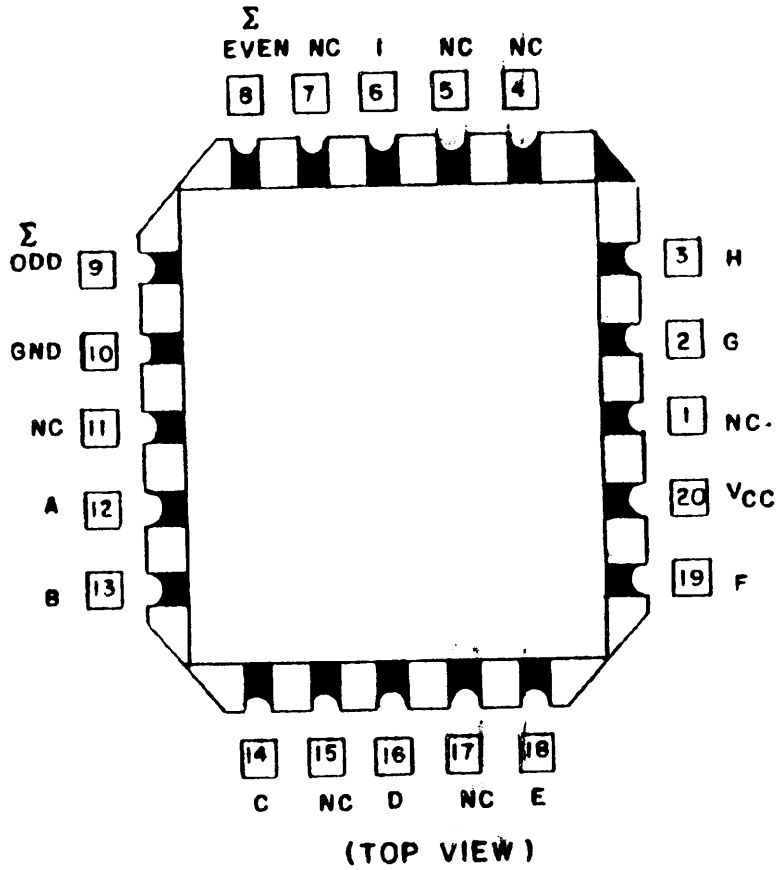


FIGURE 1. Terminal connections.

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Device type 01

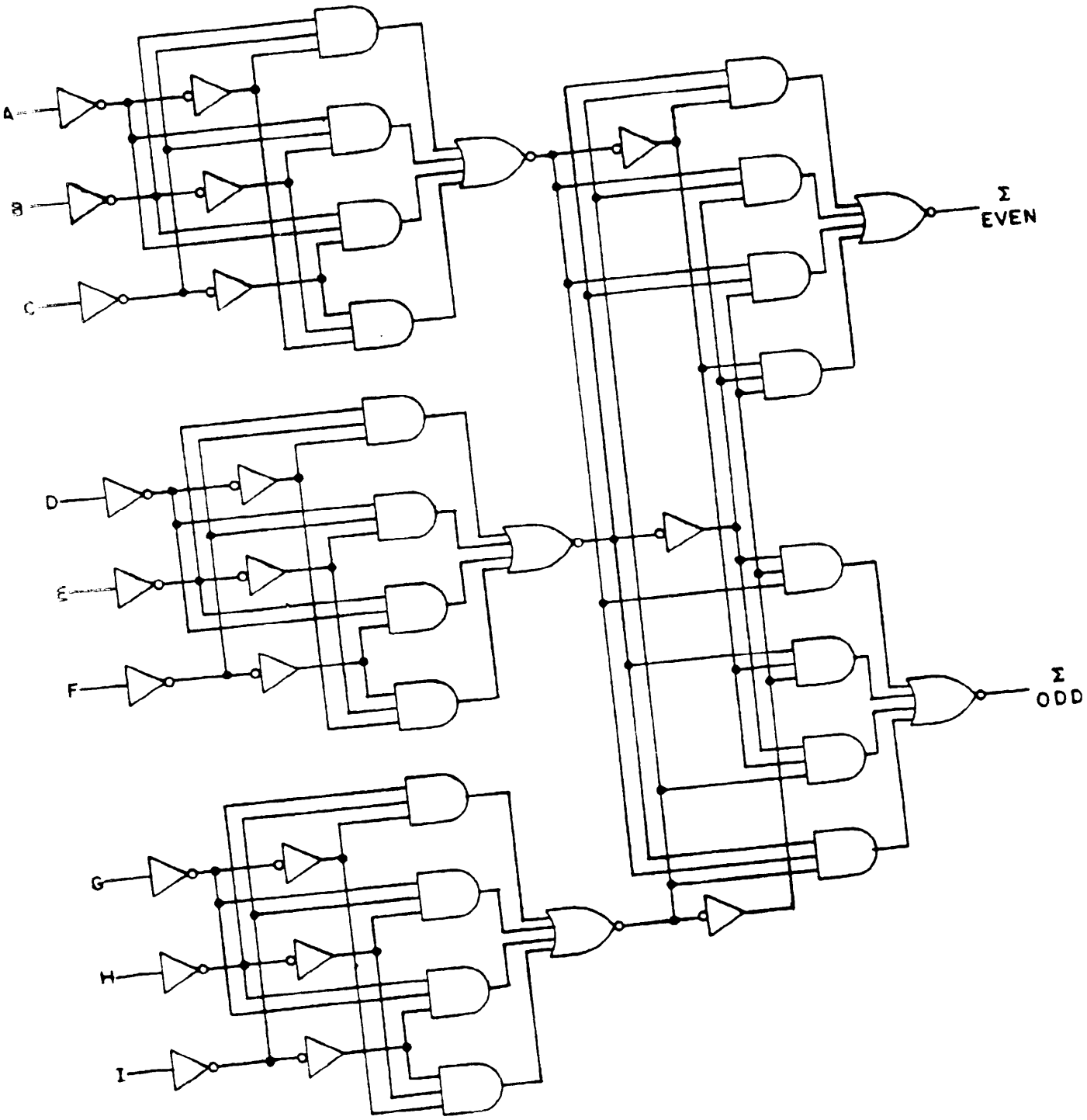


FIGURE 2. Logic diagram.

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Device type 01

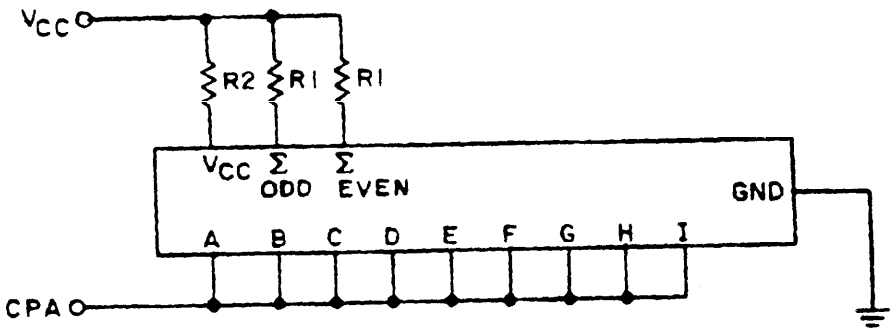
Number of inputs A thru I that are high	Outputs	
	Σ Even	Σ Odd
0,2,4,6,8	H	L
1,3,5,7,9	L	H

H = High level, L = Low level

FIGURE 3. Truth table.

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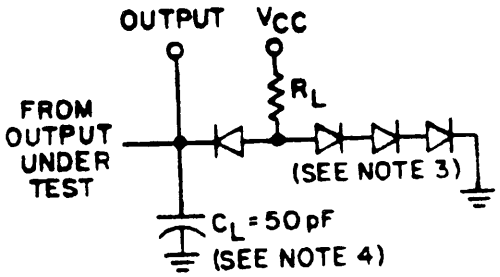
Device type 01



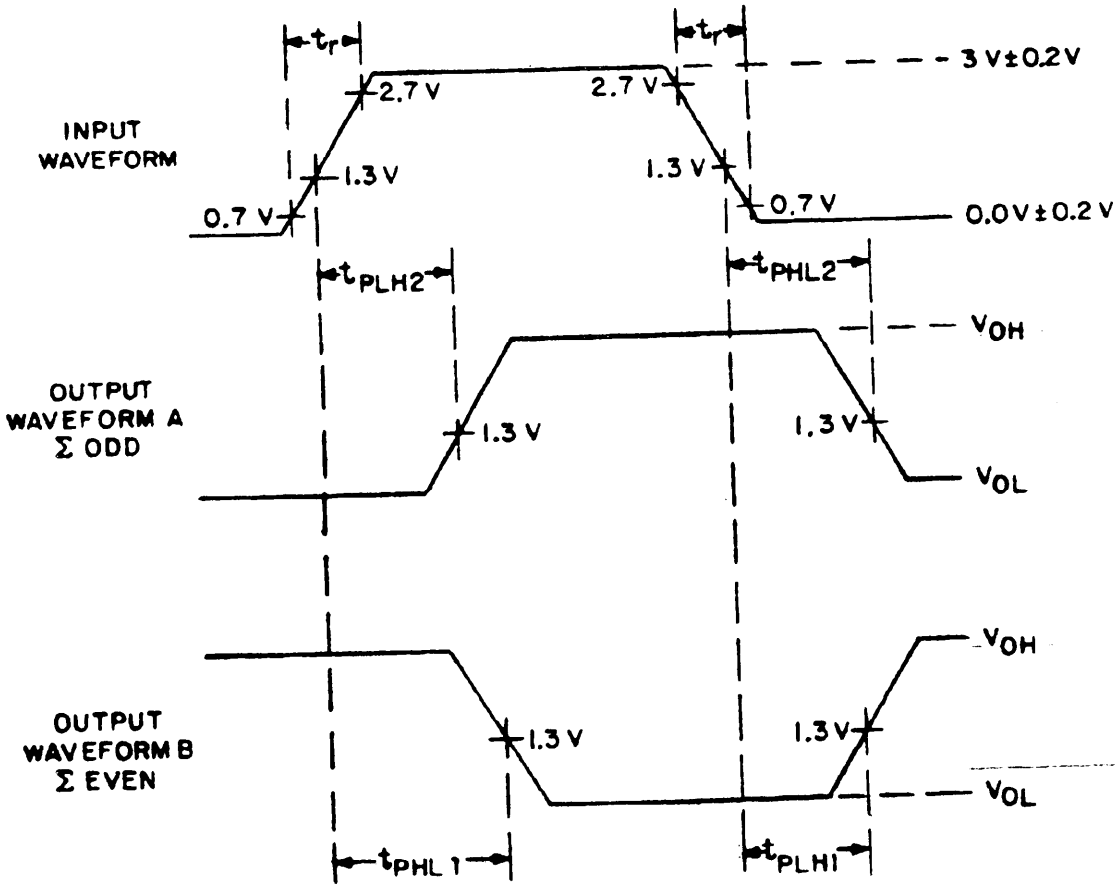
NOTES:
1. CPA = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 1 kΩ ±5%; R2 = 27Ω ±5%.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 4. Burn-in and life test circuit.

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LOAD FOR OUTPUT UNDER TEST



NOTES:
1. Input pulse characteristics: PRR ≤ 1.0 MHz, t_r ≤ 15 ns, and t_f ≤ 6 ns.
2. C_L = 50 pF ±10% including probe and jig capacitance.
3. R_L = 2.0 kΩ ±5%. All diodes are 1N3064 or 1N916.
4. Load circuit on a given output is only required where the specific test in table III indicates "OUT" on that output.

FIGURE 5. Switching test circuit and waveforms for device type 01.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases K,B,C,D Case X 1/ Test no.	1 G	2 N	3 NC	4 T	5 SE	6 SO	7 GND	8 A	9 B	10 C	11 D	12 E	13 F	14 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{IH}	3006	1	0.7 V	0.7 V		0.7 V	-0.4 mA		GND	0.7 V	0.7 V	0.7 V	0.7 V	0.7 V	0.7 V	4.5 V	SE	2.5		V
		3006	2	"	"		"		0.4 mA	"	2.0 V	"	"	"	"	"	"	SO	2.5		"
	V_{OL}	3007	3	"	"		"	4 mA		"	2.0 V	"	"	"	"	"	"	SE		0.4	"
		3007	4	"	"		"		4 mA	"	0.7 V	"	"	"	"	"	"	SO		0.4	"
	V_{IC}		5	-18 mA														G		-1.5	"
			6		-18 mA													H		"	"
			7				-18 mA											I		"	"
			8								-18 mA							A		"	"
			9									-18 mA						B		"	"
			10										-18 mA					C		"	"
			11											-18 mA				D		"	"
			12												-18 mA			E		"	"
			13													-18 mA		F		"	"
	I_{IL}	3009	14	0.4 V													5.5 V	G	2/	2/	μA
		"	15		0.4 V												"	H	"	"	"
		"	16				0.4 V										"	I	"	"	"
		"	17								0.4 V						"	A	"	"	"
		"	18									0.4 V					"	B	"	"	"
		"	19										0.4 V				"	C	"	"	"
		"	20											0.4 V			"	D	"	"	"
		"	21												0.4 V		"	E	"	"	"
		"	22													0.4 V	"	F	"	"	"
	I_{IH1}	3010	23	2.7 V														G		20	μA
		"	24		2.7 V												"	H	"	"	"
		"	25				2.7 V										"	I	"	"	"
		"	26								2.7 V						"	A	"	"	"
		"	27									2.7 V					"	B	"	"	"
		"	28										2.7 V				"	C	"	"	"
		"	29											2.7 V			"	D	"	"	"
		"	30												2.7 V		"	E	"	"	"
		"	31													2.7 V	"	F	"	"	"
	I_{IH2}	"	32	5.5 V														G		100	μA
		"	33		5.5 V												"	H	"	"	"
		"	34				5.5 V										"	I	"	"	"
		"	35								5.5 V						"	A	"	"	"
		"	36									5.5 V					"	B	"	"	"
		"	37										5.5 V				"	C	"	"	"
		"	38											5.5 V			"	D	"	"	"
		"	39												5.5 V		"	E	"	"	"
		"	40													5.5 V	"	F	"	"	"
	I_{OS}	3011	41	GND	GND		GND	GND			GND	GND	GND	GND	GND	GND		SE	-15	-100	mA
		3011	42	"	"		"		GND		4.5 V	"	"	"	"	"		SO	-15	-100	"
	I_{CC}	3005	43	"	"		"				GND	"	"	"	"	"	"	V_{CC}		27	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case X W Test no.	1 Z G	2 J H	3 4 NC	4 8 I	5 9 SE	6 V SO	7 10 GND	8 12 A	9 13 B	10 14 C	11 15 D	12 18 E	13 19 F	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
7 1/ T_C = +25°C	Truth table tests	3014	44	B	B		B	H	L	GND	B	B	B	B	B	B	5.0 V	4/			
		"	45	A	B		"	L	H	"	"	"	"	"	"	"	"	"			
		"	46	"	A		"	H	L	"	"	"	"	"	"	"	"	"			
		"	47	"	"		A	L	H	"	"	"	"	"	"	"	"	"			
		"	48	"	"		"	H	L	"	A	"	"	"	"	"	"	"			
		"	49	"	"		"	L	H	"	"	A	"	"	"	"	"	"			
		"	50	"	"		"	H	L	"	"	"	A	"	"	"	"	"			
		"	51	"	"		"	L	H	"	"	"	"	A	"	"	"	"			
		"	52	"	"		"	H	L	"	"	"	"	"	A	"	"	"			
		"	53	"	"		"	L	H	"	"	"	"	"	"	A	"	"			
		"	54	B	"		"	H	L	"	"	"	"	"	"	"	"	"			
		"	55	"	B		"	L	H	"	"	"	"	"	"	"	"	"			
		"	56	"	"		B	H	L	"	"	"	"	"	"	"	"	"			
		"	57	"	"		"	L	H	"	B	"	"	"	"	"	"	"			
		"	58	"	"		"	H	L	"	"	B	"	"	"	"	"	"			
		"	59	"	"		"	L	H	"	"	"	B	"	"	"	"	"			
		"	60	"	"		"	H	L	"	"	"	"	B	"	"	"	"			
		"	61	"	"		"	L	H	"	"	"	"	"	B	"	"	"			
		"	62	"	"		"	H	L	"	"	"	"	"	"	B	"	"			
		"	63	"	A		"	L	H	"	"	"	"	"	"	"	"	"			
		"	64	"	B		A	"	"	"	"	"	"	"	"	"	"	"			
		"	65	"	"		B	"	"	"	A	"	"	"	"	"	"	"			
		"	66	"	"		"	"	"	"	B	A	"	"	"	"	"	"			
		"	67	"	"		"	"	"	"	"	B	A	"	"	"	"	"			
		"	68	"	"		"	"	"	"	"	"	B	A	"	"	"	"			
		"	69	"	"		"	"	"	"	"	"	B	B	A	"	"	"			
		"	70	A	A		A	"	"	"	A	A	A	A	"	A	"	"			
		"	71	"	B		A	H	L	"	"	"	"	"	"	"	"	"			
		"	72	"	A		B	"	"	"	"	"	"	"	"	"	"	"			
		"	73	"	"		A	"	"	"	B	"	"	"	"	"	"	"			
		"	74	"	"		"	"	"	"	A	B	A	"	"	"	"	"			
		"	75	"	"		"	"	"	"	"	A	B	"	"	"	"	"			
		"	76	"	"		"	"	"	"	"	"	A	B	"	"	"	"			
		"	77	"	"		"	"	"	"	"	"	"	A	B	"	"	"			
		"	78	"	"		"	L	H	"	"	"	"	A	A	"	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																				
9 T_C = +25°C	t_{PLH1}	3003 Fig. 5	79	GND	GND		GND	OUT		GND	IN	GND	GND	GND	GND	GND	5.0 V	A to SE	2	55	ns
		"	80	"	"		"	"		"	GND	IN	GND	"	"	"	"	B to SE	"	"	"
		"	81	"	"		"	"		"	"	GND	IN	"	"	"	"	C to SE	"	"	"
		"	82	"	"		"	"		"	"	"	GND	IN	"	"	"	D to SE	"	"	"
		"	83	"	"		"	"		"	"	"	"	GND	IN	"	"	E to SE	"	"	"
		"	84	"	"		"	"		"	"	"	"	"	GND	"	"	F to SE	"	"	"
		"	85	IN	"		"	"		"	"	"	"	"	"	IN	"	G to SE	"	"	"
		"	86	GND	IN		"	"		"	"	"	"	"	"	GND	"	H to SE	"	"	"
		"	87	"	GND		IN	"		"	"	"	"	"	"	"	"	I to SE	"	"	"
	t_{PHL1}	"	88	"	"		GND	"		"	IN	"	"	"	"	"	"	A to SE	"	50	"
		"	89	"	"		"	"		"	GND	IN	"	"	"	"	"	B to SE	"	"	"
		"	90	"	"		"	"		"	"	GND	IN	"	"	"	"	C to SE	"	"	"
		"	91	"	"		"	"		"	"	"	GND	IN	"	"	"	D to SE	"	"	"
		"	92	"	"		"	"		"	"	"	"	GND	IN	"	"	E to SE	"	"	"
		"	93	"	"		"	"		"	"	"	"	"	GND	"	"	F to SE	"	"	"
		"	94	IN	"		"	"		"	"	"	"	"	"	IN	"	G to SE	"	"	"
		"	95	GND	IN		"	"		"	"	"	"	"	"	GND	"	H to SE	"	"	"
		"	96	GND	GND		IN	"		"	"	"	"	"	"	"	"	I to SE	"	"	"

See footnotes at end of table.

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Table III. Group A inspection for device type 01 - Continued.

Terminal conditions (pins not designated may be high ≥2.0 V, or low ≤0.7 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,E,D Case X 1/	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Test no.	2	3	4	5	6	9	10	12	13	14	16	18	19	20		Min	Max	
				S	H	NC	Y	SE	SO	GND	A	B	C	D	E	F	V_{CC}				
9 T_C = +25°C	t_{PLH2}	3003 Fig. 5	97	GND	GND		GND		OUT	GND	IN	GND	GND	GND	GND	GND	5.0 V	A to SO	2	40	ns
			98	"	"		"		"	"	GND	IN	GND	"	"	"	"	IB to SO	"	"	"
			99	"	"		"		"	"	"	GND	IN	"	"	"	"	IC to SO	"	"	"
			100	"	"		"		"	"	"	"	GND	IN	"	"	"	ID to SO	"	"	"
			101	"	"		"		"	"	"	"	"	GND	IN	"	"	IE to SO	"	"	"
			102	"	"		"		"	"	"	"	"	"	GND	IN	"	IF to SO	"	"	"
			103	IN	"		"		"	"	"	"	"	"	"	GND	"	IG to SO	"	"	"
			104	GND	IN		"		"	"	"	"	"	"	"	"	"	IH to SO	"	"	"
			105	"	GND		IN		"	"	"	"	"	"	"	"	"	II to SO	"	"	"
	t_{PHL2}	"	106	"	"		GND		"	"	IN	"	"	"	"	"	"	IA to SO	"	55	"
		"	107	"	"		"		"	"	GND	IN	"	"	"	"	"	IB to SO	"	"	"
		"	108	"	"		"		"	"	"	GND	IN	"	"	"	"	IC to SO	"	"	"
		"	109	"	"		"		"	"	"	"	GND	IN	"	"	"	ID to SO	"	"	"
		"	110	"	"		"		"	"	"	"	"	GND	IN	"	"	IE to SO	"	"	"
		"	111	"	"		"		"	"	"	"	"	"	GND	IN	"	IF to SO	"	"	"
		"	112	IN	"		"		"	"	"	"	"	"	"	GND	"	IG to SO	"	"	"
		"	113	GND	IN		"		"	"	"	"	"	"	"	"	"	IH to SO	"	"	"
		"	114	GND	GND		IN		"	"	"	"	"	"	"	"	"	II to SO	"	"	"
10 T_C = +125°C	t_{PLH1}	Same tests and terminal conditions as subgroup 9, except T_C = +125°C and limits are as shown.																	"	72	"
	t_{PHL1}																		"	65	"
	t_{PLH2}																		"	52	"
	t_{PHL2}																		"	72	"
11	Same tests, terminal conditions, and limits as subgroup 10, except T_C = -55°C.																				

1/ Pins not referenced are N/C.

2/ I_{IL} limits in μA are as follows:

Min/max limits for CKT	
A	B
-120/-360	-130/-370

3/ A ≥2.0 V and B ≤0.7 V.

4/ Output voltages shall be either

a. H @ -2.5 V minimum and L @ +0.4 V maximum when using a high speed checker double comparator, or

b. H ≥1.5 V and L ≤1.5 V when using a high speed checker single comparator.

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b. Interim and final electrical tests shall be as specified in table II, except interim electrical tests prior to burn-in are optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 tests after cooldown as the final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical paramters shall be as specified in table II.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical test parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging of microcircuits shall be in accordance with MIL-M-38510.

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NOTES

. Notes. The notes specified in MIL-M-38510 are applicable to this ification.

2 Intended use. Microcircuits conforming to this specification are intended for inal equipment design applications and logistic support of existing equipment.

3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable, these requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Electrical ground (common terminal)
I_{IN}	Current flowing into an input terminal
V_{IN}	Voltage level at an input terminal

6.5 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54LS280

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6.7 Manufacturers' designations. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuits	
	A	B
	Texas Instruments	Motorola Inc.
01	X	X

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - EC, SH, OS
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/329A	MICROCIRCUIT, DIGITAL, LOW-POWER SCHOTTKY TTL, PARITY CHECKER, MONOLITHIC SILICON

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS (Street, City, State, ZIP Code)

4. TYPE OF ORGANIZATION (Mark one)

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER (Specify): _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional

b. WORK TELEPHONE NUMBER (Include Area Code) – Optional

c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional

8. DATE OF SUBMISSION (YYMMDD)

DD FORM 82 MAR 1426 **PREVIOUS EDITION IS OBSOLETE.**

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